

NATCHITOCHES, LA, USA

WMO: 722319

Lat: 31.736N

Lon: 93.099W

Elev: 121

StdP: 14.63

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 53943

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.3	29.7	16.9	13.0	34.2	21.3	16.1	37.7	20.8	50.8	18.8	51.5	2.7	0	0.377

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.9	99.4	79.4	97.2	78.9	95.1	78.6	83.4	92.6	81.9	90.9	80.9	89.9	5.5	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.5	164.4	86.4	79.5	153.6	85.1	78.9	150.6	84.7	47.3	92.8	45.6	92.1	44.5	89.5	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years		DB	WB
Min	Max			Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.1	15.8	12.9		21.6	102.0	4.3	3.7	18.5	104.7	16.0	106.8	13.6	108.9	10.5	111.6	
				20.3	83.4	4.1	1.9	17.3	84.8	14.9	85.9	12.6	86.9	9.6	88.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	67.0	48.6	52.0	59.8	66.2	75.1	81.1	82.6	83.6	78.3	68.0	57.2	50.7
	DBStd	14.60	9.70	9.88	8.99	7.21	6.63	3.89	3.33	3.73	4.79	8.16	9.08	9.28
	HDD50	394	147	85	22	2	0	0	0	0	0	1	32	105
	HDD65	1969	516	379	208	70	9	0	0	0	0	66	265	456
	CDD50	6604	102	141	327	487	777	932	1011	1042	850	560	248	127
	CDD65	2702	6	14	48	106	321	482	546	577	400	160	30	12
	CDH74	27811	18	76	367	954	3282	5081	5912	6469	3821	1551	227	53
	CDH80	13347	0	6	58	222	1490	2579	3031	3495	1840	588	35	3
Wind	WSAvg	4.8	5.8	6.3	6.0	6.2	5.2	4.0	3.3	3.4	3.6	4.1	4.8	5.5
Precipitation	PrecAvg	53.0	4.6	4.7	4.6	4.2	5.8	4.3	3.5	3.4	3.4	3.9	4.3	6.0
	PrecMax	81.7	13.7	11.3	8.8	14.1	18.3	15.9	7.8	7.2	7.6	11.9	14.6	21.8
	PrecMin	32.0	0.6	1.4	0.8	0.3	1.1	0.7	0.2	0.2	0.6	0.0	0.6	0.8
	PrecStd	12.4	3.2	2.7	2.2	3.3	4.0	3.3	2.2	1.9	1.8	3.0	2.9	3.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.8	80.8	84.8	88.7	99.6	102.0	100.4	102.3	98.0	92.9	84.0	79.3
		MCWB	66.4	67.9	69.9	73.2	77.7	79.6	79.7	79.7	76.0	73.8	71.3	70.7
	2%	DB	72.7	76.6	81.4	85.6	93.4	98.0	98.5	99.5	95.1	89.9	80.5	74.7
		MCWB	64.3	66.3	69.5	72.8	75.3	79.5	79.9	79.5	76.8	73.4	69.4	67.9
	5%	DB	69.8	72.4	78.7	82.1	90.7	94.9	95.4	97.2	92.8	86.7	76.6	71.7
		MCWB	63.0	65.4	67.3	69.5	75.3	79.3	79.3	79.3	76.6	71.7	67.2	66.8
	10%	DB	65.6	68.8	74.7	80.1	88.3	91.4	93.0	94.9	90.3	82.4	72.5	67.9
		MCWB	59.6	63.4	65.4	68.8	74.9	78.5	79.0	78.9	75.8	70.4	66.1	64.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.4	71.2	74.3	76.6	80.9	84.9	84.5	85.6	82.8	78.9	73.6	72.9
		MCDB	73.6	76.2	80.6	82.7	87.5	90.1	90.7	93.5	88.9	84.8	79.2	77.0
	2%	WB	67.4	69.1	71.9	74.7	78.8	82.2	82.5	83.8	80.5	76.8	72.0	70.2
		MCDB	70.5	73.2	78.2	81.6	88.8	90.6	91.6	93.9	87.9	84.8	77.1	73.3
	5%	WB	64.3	66.6	69.8	73.2	77.5	81.1	81.4	82.1	79.0	74.9	70.1	67.9
		MCDB	67.7	71.3	75.0	79.4	87.6	90.4	90.5	91.5	87.2	82.8	74.2	71.0
	10%	WB	61.5	64.1	67.9	71.7	76.4	79.9	80.4	80.7	77.7	73.0	67.4	64.1
		MCDB	64.4	67.9	73.4	77.3	86.1	89.2	89.5	89.7	85.8	79.2	71.4	66.5
Mean Daily Temperature Range	5% DB	MDBR	21.0	20.4	22.6	23.0	22.5	20.8	20.5	21.9	22.1	24.6	23.4	20.6
		MCDBR	25.2	24.0	24.6	25.1	25.0	25.2	23.9	24.9	24.9	27.1	25.6	23.4
		MCWBR	16.8	15.1	13.4	12.4	9.6	9.0	7.8	7.8	8.8	11.8	15.8	16.5
	5% WB	MCDBR	20.3	20.4	20.6	20.7	21.2	21.9	20.9	22.4	20.6	20.8	20.5	20.2
		MCWBR	16.1	14.5	13.3	12.0	9.9	9.8	8.7	9.0	8.9	10.4	14.4	16.0
Clear-Sky Solar Irradiance	taub	0.331	0.348	0.370	0.408	0.439	0.475	0.490	0.488	0.436	0.368	0.355	0.339	
	taud	2.518	2.471	2.408	2.307	2.269	2.208	2.198	2.200	2.316	2.480	2.468	2.494	
	Ebn at Noon	284	289	289	280	271	260	255	255	265	278	272	274	
	Edh at Noon	27	31	36	41	43	46	46	46	39	31	29	27	
All-Sky Solar Radiation	RadAvg	867	981	1375	1699	1885	1947	1942	1833	1592	1324	980	757	
	RadStd	82	169	114	122	130	146	148	160	148	184	102	91	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+185	+124

Nomenclature: See separate page